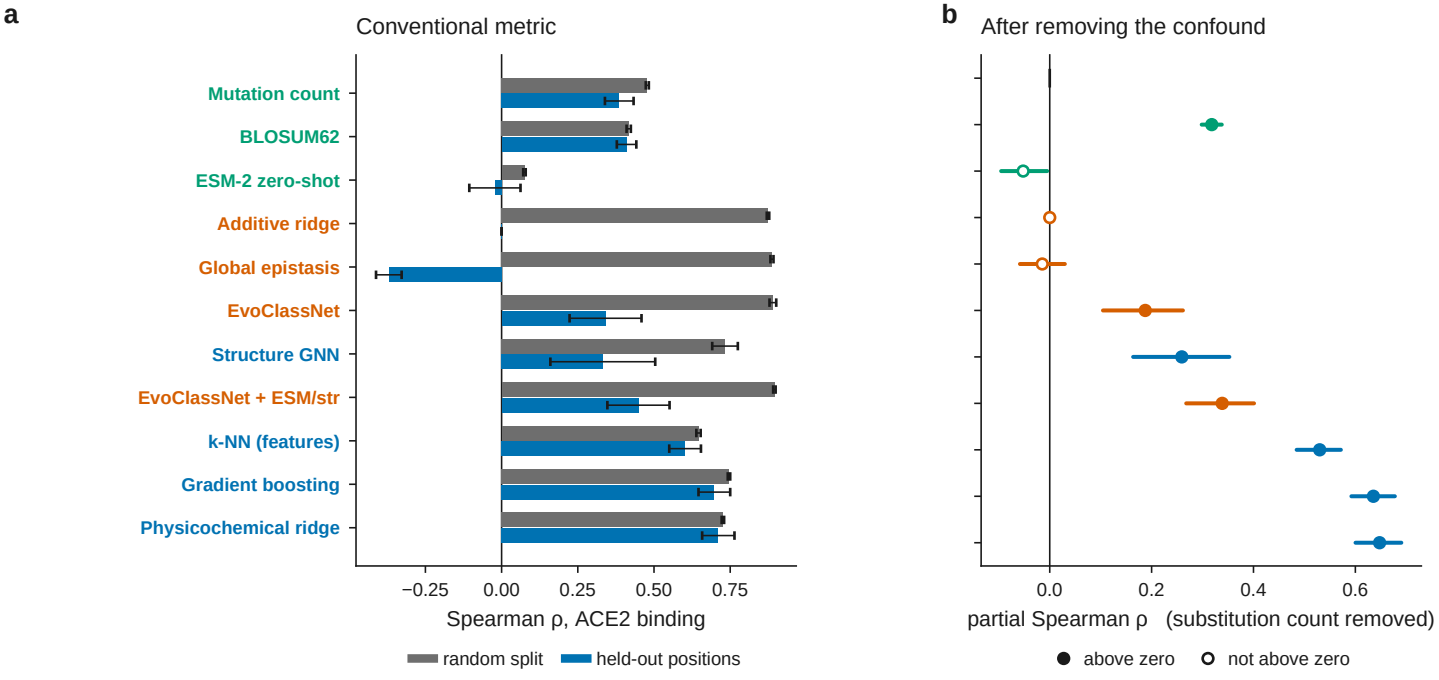
a
Conventional metric
Mutation count
BLOSUM62
ESM-2 zero-shot
Additive ridge
Global epistasis
EvoClassNet
Structure GNN
EvoClassNet + ESM/str
k-NN (features)
Gradient boosting
Physicochemical ridge
−0.25
0.00
0.25
0.50
0.75
Spearman ρ, ACE2 binding
random split
held-out positions
b
After removing the confound
0.0
0.2
0.4
0.6
partial Spearman ρ (substitution count removed)
above zero
not above zero